

# Uday Tyagi

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## Education

### Cornell University

Master's in Computer Science, GPA 3.8/4.0

Ithaca, NY

Jan. 2026 – Dec. 2026

### University of Wisconsin–Madison

Bachelor's in Computer Science, GPA 3.8/4.0

Madison, WI

May 2023 – Dec. 2025

## Professional Experience

### KLA

Software Engineer

Remote

May 2025 – Present

- Fine-tuned an LLM-powered AI agent for the company's DART web application and extended it with a custom MCP server, giving engineers natural-language, auditable tool-calling access to defect-inspection workflows
- Built visual regression and UI-performance monitoring pipelines for HPC automation workflows using Grafana Faro, reducing debugging time in large-scale wafer inspection systems

### Foresee Health

Software Engineering Intern

San Francisco, CA

Sept. 2024 – Jan. 2025

- Engineered an authentication system (AWS Amplify, Lambda) and designed PostgreSQL schemas for a microservices architecture handling large customer datasets in the invoice application

## Research

### Researcher – LAISR (Long-Term AI Safety Research)

Ithaca, NY

Cornell University, advised by Prof. Lionel Levine

- Constitutional Drift: built an Inspect AI evaluation harness testing whether an LLM's values stay stable when it repeatedly rewrites its own guiding document, scoring outputs across 9 value dimensions and checking whether the model's stated values actually match its behavior
- Designed the harness with two collaborators across 4 experimental protocols (self-editing, document curation, multi-model deliberation, fixed-point convergence), with paraphrase/random controls, cross-lab model panels, and permutation-based statistical validation

### Research Engineer – NASA USRC (Project Orion)

Ithaca, NY

Cornell University, advised by Mehrnaz Sabet

- Trained a diffusion-based motion planner replacing ORCA for real-time multi-drone collision avoidance: DiT architecture generating velocity command sequences from ego and neighbor states, with DPM-Solver++ inference optimization and low-temperature reranking, reaching ~80% mean scenario completion vs. ORCA's 59%
- Built the surrounding deconfliction stack: a ROS 2 planner plugin, scenario-based evaluation pipelines with behavioral analytics, and a ~200k-trajectory training dataset across six collision scenarios

## Projects

### NLA Steering: Text-Driven Behavioral Steering of LLMs | Python, PyTorch, HuggingFace, Qwen2.5-7B

- Used Natural Language Autoencoders (Fraser-Taliente et al., 2026) to generate steering vectors for Qwen2.5-7B from hand-written text descriptions alone — no training data, no activation extraction, no labeled examples; vectors reliably shift model behavior at inference time across multiple concepts (sycophancy, misalignment, persona)
- Running follow-on experiments ablating which parts of the text format carry the steering signal, and benchmarking the vectors as zero-shot behavioral probes against ground-truth activation-derived vectors

### SMARAG: Multi-Agent RAG System for ESG Carbon Reporting | Python, LangChain, GRPO, Prof. Oliver Gao Lab

- Built a multi-agent RAG system for automated ESG carbon reporting in Prof. Oliver Gao's lab, using LangChain orchestration for multi-agent coordination
- Applied GRPO-based reinforcement learning to train the retrieval/generation agents for factual grounding, rewarding citation accuracy and penalizing hallucinated claims in regulatory filings, benchmarked against SFT, DPO, and PPO baselines

### Sessional: AI-Powered Between-Session Patient Triage | Next.js, FastAPI, PostgreSQL, Hume AI, Gemini

- Built a full-stack clinical support app that lets patients submit async voice check-ins and gives clinicians structured, triage-ready briefs, with a Next.js frontend and a FastAPI/PostgreSQL backend handling JWT auth, role-based access, and audio ingestion
- Designed an AI signal pipeline combining Hume AI prosody/sentiment extraction with Gemini-based structured brief synthesis to surface "divergence moments" (neutral language masking elevated vocal distress), with post-generation guardrails enforcing evidence-grounded, hallucination-resistant output

## Technical Skills

**Languages:** Python, Java, C/C++, JavaScript, TypeScript

**AI/ML:** PyTorch, HuggingFace, RL(GRPO, PPO, DPO), mechanistic interpretability (probes, steering vectors, SAEs, activation patching), fine-tuning (LoRA/PEFT), evals, RAG, diffusion models, LangChain, MCP, inference optimization

**Tools & Infra:** Docker, Kubernetes, AWS (Amplify, Lambda), GCP, PostgreSQL, Kafka, Spark, React, Node.js